

CCDEVAP Term 3, AY 2025 – 2026

Machine Project – Project Specifications Document

Groupings: At most 4 members in a group

Online Airline Ticketing System

Project Overview

You are tasked to design and implement an **Online Airline Ticketing System**. The system should allow passengers to search for flights, book tickets, select optional packages (meals, seat selection, baggage), and manage reservations (modify/cancel).

The project is divided into **3 milestones**. Each milestone builds upon the previous one and introduces additional tools and requirements.

Milestone 1 — Client-Side UI Design (HTML, CSS, JavaScript, jQuery, Bootstrap)

Updated Milestone 1 — Client-Side UI Design (Enhanced Intermediate-Level Specification)

Milestone 1 — Interactive Client-Side Airline Booking System

Technologies: HTML, CSS, JavaScript, jQuery, Bootstrap

Goal:

Develop a responsive and interactive airline booking web application prototype focusing on modern UI/UX practices and application-style interfaces using only client-side technologies.

The milestone should simulate a realistic airline booking experience with interactive controls, responsive layouts, reusable UI components, and dynamic behavior using JavaScript and jQuery.

Functional Requirements

1. Application Navigation & Layout

Main Navigation Bar

Create a responsive navigation bar present on all pages.

Navigation links:

- Home / Dashboard
- Search Flights
- Book Flight
- My Reservations
- User Profile
- Admin Panel

Navigation Requirements

- Sticky/fixed top navigation
- Responsive mobile hamburger menu
- Active page highlighting
- Dropdown user menu containing:
 - Profile
 - Settings
 - Logout

UI/UX Expectations

- Consistent branding and color theme
- Use icons where appropriate
- Smooth hover transitions
- Proper spacing and alignment

- **Mobile-responsive behavior**
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2. Home / Dashboard Page

Create an application-style landing dashboard instead of a static homepage.

Required Components

- **Hero banner with airline branding**
- **Flight search quick widget**
- **Promotional cards/carousel**
- **Recently viewed or suggested flights section**
- **Flight statistics cards:**
 - **Available flights**
 - **Active bookings**
 - **Popular destinations**

Interactive Features

- **Bootstrap carousel**
 - **Animated cards on hover**
 - **Dropdown destination filters**
 - **Search auto-suggestion simulation using jQuery**
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3. Flight Search Page

Flight Search Interface

Search Form Requirements

Use modern form design with:

- **Floating labels or grouped form controls**
- **Input validation feedback**

- **Responsive grid layout**

Required Inputs

- **Origin airport**
 - **Dropdown/select component**
- **Destination airport**
 - **Dropdown/select component**
- **Trip type**
 - **One-way / Round-trip toggle**
- **Departure date picker**
- **Return date picker**
- **Passenger selector dropdown:**
 - **Adults**
 - **Children**
 - **Infants**
- **Cabin class dropdown:**
 - **Economy**
 - **Premium Economy**
 - **Business**
 - **First Class**

Advanced Search Section

Collapsible/accordion panel containing:

- **Preferred airline dropdown**
- **Direct flights only checkbox**
- **Flexible dates toggle**

- **Price range slider**
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Flight Results Interface

Results Layout

Display results using:

- **Cards OR table with expandable rows**

Each result must include:

- **Airline logo**
- **Flight number**
- **Departure and arrival times**
- **Flight duration**
- **Number of layovers**
- **Ticket price**
- **Remaining seat count**
- **Book button**

Interactive Features

- **Sort dropdown:**
 - **Price**
 - **Departure time**
 - **Duration**
- **Filter sidebar:**
 - **Airlines**
 - **Price ranges**
 - **Stops**

- **Departure schedules**
- **Expandable “View Details” section**
- **Hover effects and transitions**

Dummy Data

Use at least:

- **8–10 sample flights**
 - **Multiple airlines**
 - **Different pricing categories**
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4. Booking / Reservation Page

Passenger Information Form

Required Fields

- **Full Name**
- **Email Address**
- **Contact Number**
- **Passport Number**
- **Nationality dropdown**
- **Date of Birth**
- **Gender selection**
- **Emergency Contact**

UI/UX Requirements

- **Multi-section card layout**
- **Inline validation messages**
- **Progress indicator/stepper**

Optional Package Selection

Meal Selection

Interactive dropdown/select menu:

- **Standard**
- **Vegetarian**
- **Vegan**
- **Halal**
- **Kosher**
- **Gluten-Free**

Display:

- **Meal descriptions**
- **Additional cost updates dynamically**

Seat Selection System

Requirements

Implement an interactive seat map using:

- **Bootstrap Grid**
- **jQuery click events**

Features

- **Different seat statuses:**
 - **Available**
 - **Selected**
 - **Occupied**

- Premium
- Legend section
- Real-time seat selection updates
- Prevent selecting occupied seats
- Tooltip or modal showing seat details

UI Enhancements

- Hover highlights
 - Color-coded seats
 - Selected seat animations
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Extra Services Section

Provide interactive controls for:

- Additional baggage quantity selector
- Priority boarding toggle
- Travel insurance checkbox
- Lounge access option

Each option must:

- Dynamically affect total pricing
 - Show descriptive tooltips/modals
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5. Booking Summary Sidebar

Create a fixed or sticky booking summary panel.

Required Information

- Selected flight details

- Passenger count
- Selected seat(s)
- Meal package
- Extra services
- Taxes and fees
- Grand total

Interactive Features

- Real-time price calculation
 - Dynamic updates without page refresh
 - Expand/collapse breakdown section
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6. My Reservations Page

Reservation Management Interface

Display reservations using:

- Cards
OR
- Data table with pagination

Reservation Information

- Booking reference
- Passenger name
- Flight route
- Seat number
- Booking status
- Total price

Required Features

- Search bar
- Sort dropdown
- Filter by status
- View details modal
- Cancel booking button
- Edit reservation button (UI simulation only)

Interactive UX

- Status badges
- Expandable reservation cards
- Confirmation modals

Use dummy/static data only.

7. User Profile Page

Required Sections

- Personal information
- Saved passengers
- Payment method UI mockup
- Notification preferences
- Travel history

Interactive Components

- Editable forms
- Toggle switches
- Tabs or accordion sections
- Profile image placeholder upload UI

8. Admin Panel UI (Frontend Only)

Create admin interface mockups.

Required Admin Pages

Flight Management

- **Flight table**
- **Add flight modal**
- **Edit flight modal**
- **Delete confirmation modal**

User Management

- **User table**
- **Search/filter controls**
- **Status indicators**

Reservation Management

- **Reservation table**
- **Booking status dropdown**
- **Action buttons**

Dashboard Analytics

Display sample charts/cards for:

- **Total bookings**
- **Revenue**
- **Popular destinations**

9. Required Interactive UI Components

Students MUST implement the following:

- **Dropdown menus**
 - **Modal windows**
 - **Accordions/collapsible sections**
 - **Tooltips/popovers**
 - **Dynamic tables**
 - **Tabs**
 - **Responsive navigation**
 - **Progress bars/steppers**
 - **Interactive cards**
 - **Toast or alert notifications**
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10. JavaScript & jQuery Requirements

Students must demonstrate:

- **DOM manipulation**
 - **Event handling**
 - **Dynamic content updates**
 - **Form validation**
 - **Conditional rendering**
 - **UI state changes**
 - **Dynamic price computation**
 - **Interactive filtering and sorting**
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11. UI/UX Design Standards

Design Expectations

- **Modern application-style layout**
- **Consistent spacing and typography**
- **Responsive design for:**
 - **Desktop**
 - **Tablet**
 - **Mobile**
- **Accessibility-conscious design**
- **Clear visual hierarchy**

UX Requirements

- **Minimize excessive scrolling**
- **Logical user flow**
- **Immediate visual feedback**
- **Error message clarity**
- **Interactive transitions and animations**

12. Technical Constraints

Allowed Technologies

- **HTML5**
- **CSS3**
- **JavaScript**
- **jQuery**
- **Bootstrap**

NOT Allowed

- **React**
 - **Vue**
 - **Angular**
 - **Tailwind**
 - **Backend/database integration**
 - **External component frameworks**
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13. Deliverables

Students must submit:

- **Complete HTML/CSS/JS source code**
 - **Organized project folder structure**
 - **External stylesheet and JS files**
 - **Responsive implementation**
 - **Proper commenting/documentation**
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14. Suggested Pages Structure

Recommended pages:

- **index.html**
- **search.html**
- **booking.html**
- **reservations.html**
- **profile.html**
- **admin-dashboard.html**
- **admin-flights.html**
- **admin-users.html**

- **admin-reservations.html**
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15. Suggested Intermediate-Level Enhancements

Students are encouraged to implement:

- **Dark/light mode toggle**
 - **Loading spinners**
 - **Skeleton loading placeholders**
 - **Animated transitions**
 - **Simulated notifications**
 - **Multi-step booking wizard**
 - **Flight comparison modal**
 - **Recently searched flights**
 - **Favorite destinations UI**
 - **Simulated chat support widget**
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Evaluation Emphasis

Higher grades will be given to projects that demonstrate:

- **Strong UI/UX consistency**
- **Rich interactivity**
- **Clean component organization**
- **Professional application-style design**
- **Responsive implementation quality**
- **Effective use of jQuery and Bootstrap**
- **Proper code readability and maintainability**

IMPORTANT: Note that for every instance of a create feature that adds data in the database of the web application, there should be corresponding features that will allow the corresponding user to read, update, and delete the data from the database. For example, if you have a feature that allows the user to create a post, the user should also be allowed to read/view the post, update/edit the post, and delete/remove the post.

WORKING WITH GROUPMATES

For this project, you are encouraged to work in groups of at most 4 members. Make sure that each member of the group has approximately the same amount of contribution for the project. Problems with groupmates must be discussed internally within the group, and if needed, with the lecturer.

USE OF AI

As this is common core course and as stated in the course syllabus, AI should not be used in any portion of the submission of this project.